

Aliah University

Even (6th) Semester Examination, 2022-2023

Course: Civil Engineering

Year: Third Year

Subject: Environmental Engineering

Subject Code: CENUGPC18(PCC)

Max. Marks: 80

Time: 3 hrs

1. Answer any two questions: (20X2=40)

- a) Derive the equation for Logistic curve (S-Curve) method for population forecasting. Forecast the population for the year 2050 for a town from the data furnished below by i) Incremental Increase Method, ii) Geometric Increase Method: (20)

Year	Population (in thousands)
1980	90
1990	110
2000	160
2010	200

- b) Water is to be supplied to a town of population 1.5 lakh. If the water works is situated at a lower elevation of 50 metres than the water level in the source, then determine the size of the gravity main required to convey water from the source to the water work. Given, the length of the gravity main is 25 km and the per capita demand of the town is 135 lpcd. (20)
- c) Water has to be supplied to a town with one lakh population at the rate of 150 lpcd from a river 2000 m away. The difference in elevation between the lowest water level in the sump and the reservoir is 36 m. If the demand has to be supplied in 8 hrs, then determine the size of the main and the B.H.P of the pump required. Assume maximum demand is 1.5 times the average demand. Assume $f=0.0075$, velocity in the pipe is 2.4 m/sec and the efficiency of the pump is 80%. (20)

2. Answer any two questions: (20X2=40)

- a) i) State Dupuit's assumptions. (4)
ii) Derive the Dupuit's equation for determination of yield from a confined aquifer. (6)
iii) A 80 cm diameter well is being pumped at a rate of 1450 litres/minute. Measurement in a nearby test well were made at the same time as follows. At a distance of 6 m from the well being pumped, the drawdown was 7 m, and at 15 m, the drawdown was 2.5 m. The bottom of the well is 100 m below the ground water table. (a) Find out the coefficient of permeability. (b) If all the observed pts. were on the Dupuit's curve, what was the drawdown in the well during pumping? (c) What is the sp. capacity of the well? (d) What is the maximum rate at which water can be drawn from this well? (10)

- b) Calculate the head losses and the corrected flows in the various pipes of a distribution network as shown in Figure 1 below. The diameters (d) and the length of the pipes used are given against each pipe. Make use of Hardy- Cross Method with William Hazen's formula. Compute the corrected flows after one correction. (20)

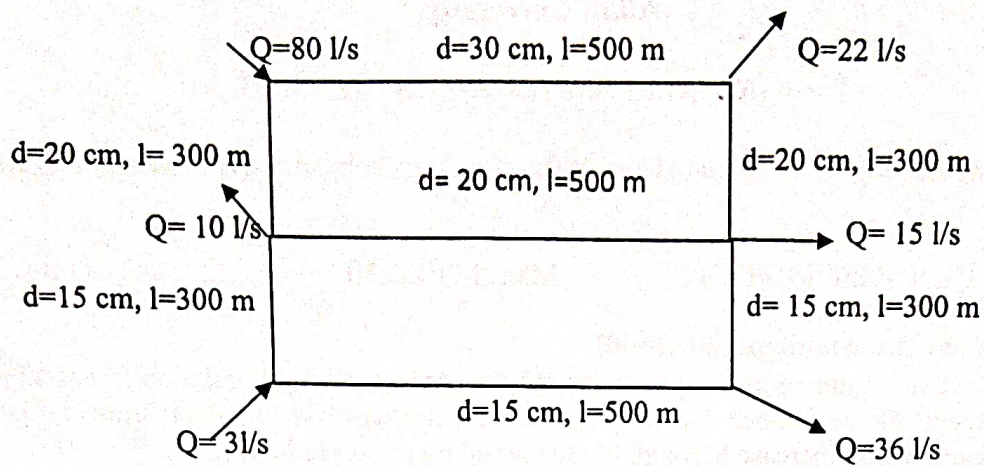


Figure 1

- c) A city with population of 2 lakhs has to be supplied with water at 150 litres per person per day. The probable hourly variation in the rate of demand is given below in Table 1:

Table 1

Period of day in hrs	Percentage of average hourly flow expected
0-1	15
1-2	15
2-3	15
3-4	20
4-5	25
5-6	40
6-7	80
7-8	120
8-9	180
9-10	220
10-11	220
11-12	150
12-13	100
13-14	80
14-15	60
15-16	110
16-17	150
17-18	180
18-19	180
19-20	160
20-21	140
21-22	80
22-23	45
23-24	15

Determine the capacity of the balancing reservoir to be provided for balancing the variable demand under a constant rate of pumping, if i) pumping is to be done for all the 24 hrs, ii) if pumping is to be done from 5 A.M. to 11 A.M. and 2 P.M. to 8 P.M. Also mention the rate of pumping in both the cases.

Solve the problem either by Mass Curve Method or by Analytical Method. (20)

ALIAH UNIVERSITY
End-Term Examination- 6th Semester
Session-2022-2023

Year: 3rd Year
Semester: 6th Semester
Subject: Foundation Engineering
Subject Code: CENUGPC19

Full Marks: 80
Time: 3hr.

Answer question no.1 and any three from the rest:

1. Answer the question briefly

[4*5=20]

- i) What is negative skin friction on pile foundation? How is it calculated for a single pile and a group of piles in clay? **3**
- ii) Which soil parameter can be taken as a measure of the degree of over-consolidation and rate of consolidation? Give a few causes of preconsolidation in soil.
- iii) Briefly state the disadvantages of direct shear test. **$3\frac{1}{2}$**
- iv) What is meant by undisturbed soil sample? List the field tests and laboratory tests for which disturbed and undisturbed soil samples are required. **$2\frac{1}{2}$**
- v) What are the basic characteristics of the failure mechanisms in general shear failure, punching shear failure and local shear failure? **3**

- 2. a) Differentiate shallow and deep foundation.
- b) What are the different components of settlement of shallow foundation?
- c) Define uniform settlement, differential settlement and angular distortion.
- d) Briefly discuss the objective of plate load test and drawback of plate load test.

[4+4+4+8=20]

- 3. a) What is the need to know the maximum pre-consolidation stress of a soil stratum? Discuss an empirical method to determine pre-consolidation stress. **$1\frac{1}{2}$** [10]
- b) A clay soil, tested in a consolidometer, showed a decrease in void ratio from 1.20 to 1.10 when the pressure was increased from 0.25 to 0.50 kgf/cm². Calculate coefficient of compressibility (a_v) and coefficient of volume compressibility (m_v). If the coefficient of consolidation (c_v) determined in the test for the given stress increment was 10 m²/year, calculate the coefficient of permeability in cm/s. If the sample tested at the site was taken from a clay layer 3.0 m in thickness, determine the consolidation settlement resulting from the given stress increment. **10** [10]

- 4. a) i) Write Coulomb's shear strength equation. **3** ii) What are the test conditions for UU test, CU test and CD test? Discussed the time required by each of the tests. **$4\frac{1}{2}$** iii) Write the assumptions made in Terzaghi's bearing capacity equation. **$1\frac{1}{2}$** [2+5+3=10]

$\hookrightarrow = 9$

b) Determine ultimate net and safe bearing capacity for a footing 2 m x 2 m on a soil with density of 1800 kg/m^3 , $\phi=15^\circ$; $C=15 \text{ kN/m}^2$, if it is laid at a depth of 1.5 m. Provided: $N_c=12.9$; $N_q=4.4$; $N_r=2.5$ (2)
[10]

5. a) Classify piles based on different criteria. (4)

b) What are the various approaches used to estimate the vertical load bearing capacity of a pile (1)

c) Write down the static pile load formula for estimating the point load resistance and skin friction resistance of a single pile installed in sand and clay. (2)

d) A 500mm x 500mm concrete pile, 15 m long, penetrates a deep uniform deposits of clay ($C_u=40 \text{ kN/m}^2$) and rests in a stratum of good bearing. Estimate the load taken by skin friction. Use adhesion factor, $\alpha=0.8$ (6)
[5+2+5+8=20]

ALIAH UNIVERSITY
Department of Civil Engineering
B.Tech/CENUGPE01/PEC-1/2023
Railway Engineering

Time Allotted: 03 Hours

Full Marks: 80

The figures in the margin indicate full marks.

*Candidates are requested to give the answer in their own words as far as practicable.
Assume reasonable values of data if not supplied. Answer of one complete question, including sub
questions must be written simultaneously.*

GROUP A
(Short answer type questions)

Answer all questions:

(10×3 =30)

1. (a) Distinguish between short and long welded rails.
(b) A B.G. track has a sleeper density of (M+6). If the track is laid with welded rails of 39 metres in length, find out the number of sleepers on the rail length.
(c) What do you mean by canting of rails?
(d) What do you mean by wind resistance?
(e) Differentiate between supported rail joints and suspended rail joints.
(f) What do you mean by rolling resistance?
(g) What do you mean by resistance due to track profile?
(h) Explain the necessity of grade compensation at curves.
(i) How do you define the superelevation?
(j) Write down the functions of points and crossing in the railway track layout?

GROUP B
(Long answer type questions)

Answer any five of the following:

(5×10 = 50)

2. Find the steepest gradient on a straight track using the following data for a train with wagons ^{with} 20.
(a) Weight of each wagon = 18 tonnes
(b) Rolling resistance of wagon = 2.5 kg/tonne
(c) Speed of the train = 50 km.p.h.
(d) Weight of locomotive with tender = 120 tonnes
(e) Tractive effort of locomotive = 12 tonnes
(f) Rolling resistance of locomotive = 3.5 kg/tonne

(10)

3. (a) Describe the functions and requirements of rails in a railway track.
 (b) Discuss the different types of rail sections used on B.G. and M.G. tracks in India with sketches. (10)
4. (a) What do you understand by negative super elevation? Explain with the sketch.
 (b) Calculate the maximum permissible speed on a curve of high-speed B.G. track having the following particulars.
 (i) Degree of curve = 1°
 (ii) Amount of super elevation = 8.0 cm
 (iii) Maximum cant deficiency for high-speed track = 10 cm
 (iv) Length of transition curve = 130 m
 (v) Maximum speed of the section likely to be sanctioned = 153 Km/h (10)
5. Explain the followings: (i) Check rails (ii) Turn out (iii) Heel block (iv) Square crossings. (10)
6. A 5° curve diverges from a 3° main curve in reverse direction in the layout of a B.G. yard. If the speed on the branch line is restricted to 35 km/h, determine the restricted speed on the main line [Cant deficiency for B.G. permitted is = 7.6 cm]. (10)
7. (i) Write the relative merits and demerits of flat-footed rails and bull-headed rails (ii) Differentiate warner signals and shunting signals (iii) Differentiate between semaphore signal and warner signal (iv) Write down the parts of turnouts with sketch. (10)

Even Semester Examination 2022-23

Subject Code: CENUGPE02

Subject Name: Surface Hydrology

B. Tech Civil Engineering: 6th Semester

Full Marks: 80 (8x10)

Time: 3 Hrs.

Attempt Any 8 Questions

1. Derive the equation for the Soil Conservation Service (SCS) for the computation of runoff.
2. Write down the GEE script for downloading the precipitation data having the latitude: 22.57°N and longitude: 88.36°E:
 - a) Daily precipitation for 2019 year
 - b) Monthly precipitation for 2020 year
3. How precipitation plays a key role in governing hydrologic cycle. Explain the different forms and types of precipitation.
4. There are five rain gauge station existing in a river catchment. The average annual rainfall value at these stations are 650mm, 700mm, 525mm, 680mm, and 590mm. Allowable percentage error in mean value of rainfall is 10%. Find the optimum number of rain gauge in the catchment?
5. Compute the mean areal rainfall of the watershed by using Thiessen Polygon Method. The details are given below:

Rain Gauge Station	A	B	C	D
Measured Rainfall (cm)	9.5	10.56	8.75	11.25
Area of enclosed polygon (sqkm)	17.35	16.80	18.50	15.0

6. Describe various hydrometeorological variables affecting hydrologic cycle.

Determine the length of rainfall record, if return period of highest rainfall of 75cm is 25 years.

7. Describe WMD and IMD guidelines for distribution of rain gauge. Describe the method for the computation of missing rainfall data.

8. The rainfall vs time data derived from mass curve of a specific rainfall event is given. Determine the depth of rainfall of the rainfall event.

Rainfall Time (hr)	0-1.5	1.5-3.0	3.0-4.5	4.5-6.0	6.0-7.5	7.5-9.0
Rainfall intensity (cm/hr)	3.75	5.50	6.25	5.05	6.50	3.75

9. Determine the runoff depth in response to a given rainfall of 20cm from a watershed of 600ha area. Take level of initial abstraction as 1.5cm.

10. Define the following terms:

- a) Relief and Relief Ration
- b) Drainage Density
- c) Watershed
- d) Infiltration
- e) Interception

77.45

MATERIAL SCIENCE & METALLURGY

MENUGOE01

Full Marks = 80

Time – 3 hours

(Answer any **FOUR**)

1. (a) Classify Engineering materials. Briefly describe their properties and applications. 8
- (b) Show that the Atomic Packing Fraction for FCC crystal is 0.74 5
- (c) What are the 14 Bravais Lattice? 7
2. (a) Find the density of Aluminium (FCC) if atomic weight of Aluminium is 26.98 and the lattice parameter be 4.049 Å 10
- (b) Determine the size of a tetrahedral void in a unit cell. 10
3. (a) Describe Corundum structure and explain why is Alumina very hard 6
- (b) Show $(10\bar{1})$; (012) , $(\bar{1}\bar{1}1)$ planes, and $[11\bar{1}]$; $[\bar{1}00]$; $[122]$ directions in a cubic unit cell 6
- (c) If a first order X-ray, having wavelength of 1.72Å, showing diffraction angle of 30° using (111) plane of a BCC crystal, Find the radius of one atom of the material forming the crystal. 8
4. (a) Explain different types of Point defects with sketches 6
- (b) What are the differences between the Edge and Screw Dislocations? Explain with sketch 8

(2)

- (c) What are the two types of dislocation motion? Compare them 6
5. (a) Deduce the expression for fraction of vacancy site (n_v/n). 12
- (b) Determine the number of vacancies per cubic meter occurring in Cu at 100°C, if Cu has an FCC lattice with lattice parameter of 0.3615 nm and 83.7 KJ/mole of energy required to create vacancy. Atomic wt. of Cu is 63.5. 8
6. (a) What are the differences between Hot working and Cold working? 5
- (b) Compare between the 'slip' and 'twinning' mode of plastic deformation. 6
- (c) Derive the expression for CRSS 5
- (d) State Schmid's Law. 2
- (e) Explain the reason for 'neck formation' in case a long crystalline sample is subjected to tensile stress. 2
7. (a) Derive Gibb's Phase Rule 7
- (b) What causes Segregation in metal casting? 4
- (c) Derive 'Lever Rule' for phase diagrams. 5
- (d) What are the 'Hume-Rothery Rules' for alloys? 4